

DSA Mock 1

Total points 16/40

The respondent's email (**maheshpatil9691@gmail.com**) was recorded on submission of this form.

✗ 25.State True or False.

*

0/1

i) Binary search is used for searching in a sorted array.

ii) The time complexity of binary search is $O(\log n)$.

☒ True, False

✗

☐ False ,True

☐ False ,False

☐ True ,True

Correct answer

☒ True ,True

- ✓ 10. What would you enter in the blank to run through all elements of the array? *1/1

```
int arr[] = {1,2,3,4};  
  
for (int i = 0; i < _____ ; i++){  
  
    System.out.println(arr[i]);  
  
}
```

- ☒ arr.length
- ☐ arr.length()
- ☐ arr.size()
- ☐ arr.size



- ✓ 17. What is the minimum number of queues required for implementing a stack? *1/1

- ☐ 1
- ☒ 2
- ☐ 3
- ☐ 4



✓ 18. What is the maximum number of edges possible in a directed graph having 8 vertices and no self-loops? *1/1

- ☐ 60
- ☐ 30
- ☒ 56
- ☐ 216



✗ 28. A full binary tree with n leaves contains *

0/1

- ☒ n - 1 nodes
- ☐ $\log_2 n$ nodes
- ☐ $2n - 1$ nodes
- ☐ 2^n nodes



Correct answer

- ☒ $2n - 1$ nodes

Name *

Mahesh Patil

✗ 7. Which of the following sorting algorithms is preferred to sort a linked list? *0/1

- ☐ Merge Sort
- ☐ Quick Sort
- ☐ Insertion Sort
- ☒ None of the Above

✗

Correct answer

- ☒ Merge Sort

✓ 35. { * 1/1
Stack S; // Say it creates an empty stack S

```
// Run while Q is not empty
while (!isEmpty(Q))
{
    // deQueue an item from Q and push the dequeued item to S
    push(&S, deQueue(Q));
}

// Run while Stack S is not empty
while (!isEmpty(&S))
{
    // Pop an item from S and enqueue the popped item to Q
    enqueue(Q, pop(&S));
}
}
```

- ☐ Removes the last from Q
- ☐ Keeps the Q same as it was before the call
- ☐ Makes Q empty
- ☒ Reverses the Q

✓

✗ 39. Consider an implementation of unsorted singly linked list. Suppose it has its representation with a head and tail pointer. Given the representation, which of the following operation can be implemented in $O(1)$ time? *0/1

- i) Insertion at the front of the linked list
- ii) Insertion at the end of the linked list
- iii) Deletion of the front node of the linked list
- iv) Deletion of the last node of the linked list

- ☐ I and II
- ☐ I and III
- ☐ I, II and III
- ☒ I, II and IV

✗

Correct answer

- ☒ I and III

✗ 5. Suppose a circular queue of capacity $(n - 1)$ elements is implemented with an array of n elements. Assume that the insertion and deletion operation are carried out using REAR and FRONT as array index variables, respectively. Initially, REAR = FRONT = 0. The conditions to detect queue full and queue empty are 0/1

- ☐ Full: $(\text{REAR} + 1) \bmod n == \text{FRONT}$, empty: $\text{REAR} == \text{FRONT}$
- ☒ Full: $(\text{REAR} + 1) \bmod n == \text{FRONT}$, empty: $(\text{FRONT} + 1) \bmod n == \text{REAR}$
- ☐ Full: $\text{REAR} == \text{FRONT}$, empty: $(\text{REAR} + 1) \bmod n == \text{FRONT}$
- ☐ Full: $(\text{FRONT} + 1) \bmod n == \text{REAR}$, empty: $\text{REAR} == \text{FRONT}$

✗

Correct answer

- ☒ Full: $(\text{REAR} + 1) \bmod n == \text{FRONT}$, empty: $\text{REAR} == \text{FRONT}$

✓ 31. Which of the following is the infix expression? *

1/1

- ☐ ABC+*
- ☒ A+B*C
- ☐ +A*BC
- ☐ None of the above



✗ 33. In a leftist heap, all the operations should be performed on? *

0/1

- ☒ left path
- ☐ centre path
- ☐ root
- ☐ right path



Correct answer

- ☒ right path

✗ 38. Which of the following option is not correct? *

0/1

- ☐ A-If the queue is implemented with a linked list, keeping track of a front pointer, Only rear pointer s will change during an insertion into an non-empty queue.
- ☒ B-Queue data structure can be used to implement least recently used (LRU) page fault algorithm and Quick short algorithm. ✗
- ☐ C-Queue data structure can be used to implement Quick short algorithm but not least recently used (LRU) page fault algorithm.
- ☐ Both (A) and (C)

Correct answer

- ☒ C-Queue data structure can be used to implement Quick short algorithm but not least recently used (LRU) page fault algorithm.

PN *

43

- ✗ 16. A hash table of length 10 uses open addressing with hash function $h(k) = k \bmod 10$, and linear probing. After inserting 6 values into an empty hash table, the table is as shown below. *0/1

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	

- ☐ 46, 42, 34, 52, 23, 33
- ☐ 34, 42, 23, 52, 33, 46
- ☐ 46, 34, 42, 23, 52, 33
- ☒ 42, 46, 33, 23, 34, 52

✗

Correct answer

- ☒ 46, 34, 42, 23, 52, 33

✗ 12. Do both of these statements create a string? *

0/1

```
String str = new String("abc");  
  
//or  
  
String str1 = "abc";
```

- ☐ Yes
- ☒ NO
- ☐ First is correct
- ☐ Second is correct

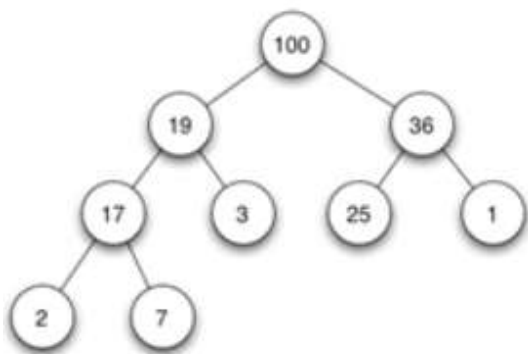
✗

Correct answer

- ☒ Yes

✓ 32. . If we implement heap as maximum heap , adding a new node of value 15 to the left most node of right subtree. What value will be at leaf nodes of the right subtree of the heap.

*1/1



- ☒ 15 and 1
- ☐ 25 and 1
- ☐ 3 and 1
- ☐ 2 and 3

✓

✓ 40. In worst case, the number of comparison need to search a singly linked list of length n for a given element is *1/1

- ☐ $\log n$
- ☐ $n/2$
- ☐ $\log_2 n - 1$
- ☒ n



✓ 1. Which of the following is a linear data structure? * 1/1

- ☒ Array
- ☐ AVL Trees
- ☐ Binary Trees
- ☐ Graphs



✗ 21. The result of prefix expression $* / b + - d a c d$, where $a = 3, b = 6, c = 1, d = 5$ is *0/1

- ☒ 0
- ☐ 5
- ☐ 10
- ☐ 8



Correct answer

- ☒ 10

- ✗ 23. Below program that takes a number as an argument, and uses a stack *0/1 S to do processing.

```
public void convert( int no ) {  
    Stack St; // For creating the empty stack  
    while ( no > 0 ) {  
        // For pushing the value of no%2 into stack St  
        push( St, no%2 );  
        no = no/2;  
    }  
    // Continuously checking whether stack is empty or not  
    while ( !isEmpty( St ) ) {  
        System.out.println( "%d ", pop( St )); // pop an element from St and print it  
    }  
}
```

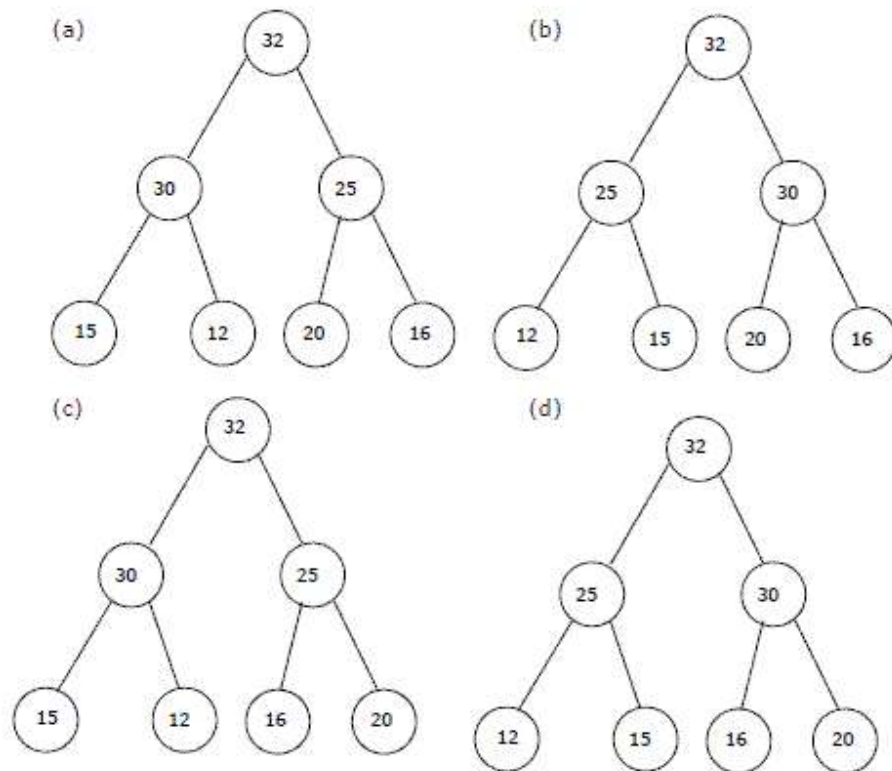
- ☒ Prints binary representation of number in reverse order ✗
- ☐ Prints binary representation of number
- ☐ Prints the value of number
- ☐ Prints the value of number in reverse order

Correct answer

- ☒ Prints binary representation of number

✗ The elements 32, 15, 20, 30, 12, 25, 16 are inserted one by one in the given order into a Max Heap. The resultant Max Heap is.

*0/1



- ☐ a
- ☐ b
- ☐ c
- ☒ d

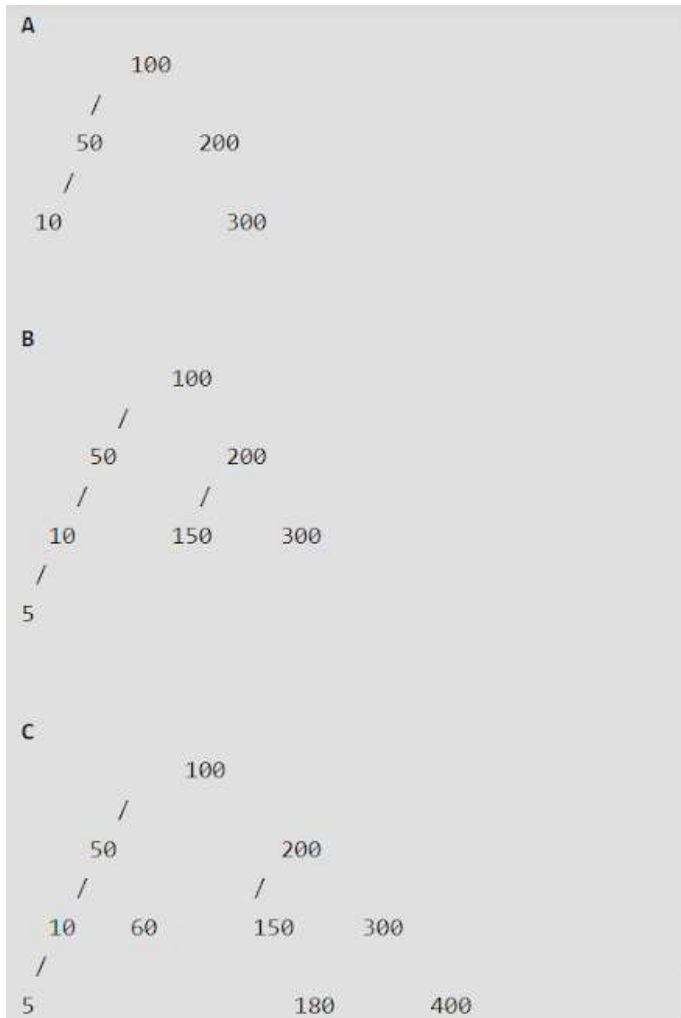
✗

Correct answer

- ☒ a

✗ 6.Which of the following is AVL Tree *

0/1



- ☒ A
- ☐ A and C
- ☐ A,B and C
- ☐ B only

Correct answer

- ☒ A and C

✗

✓ 2. Which of the following is not the type of queue? *

1/1

- ☐ Priority queue
- ☒ Single-ended queue
- ☐ Circular queue
- ☐ Ordinary queue



✗ 3.What will be the output of the following code snippet

0/1

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        solve();
    }

    public static void solve() {
        int n = 24;
        int l = 0, r = 100, ans = n;
        while (l <= r) {
            int mid = (l + r) / 2;
            if (mid * mid <= n) {
                ans = mid;
                l = mid + 1;
            } else {
                r = mid - 1;
            }
        }
        System.out.println(ans);
    }
}
```

- ☒ 5
- ☐ 4
- ☐ 6
- ☐ 3

✗

Correct answer

- ☒ 4

✓ 20. In a circular queue the value of r will be *

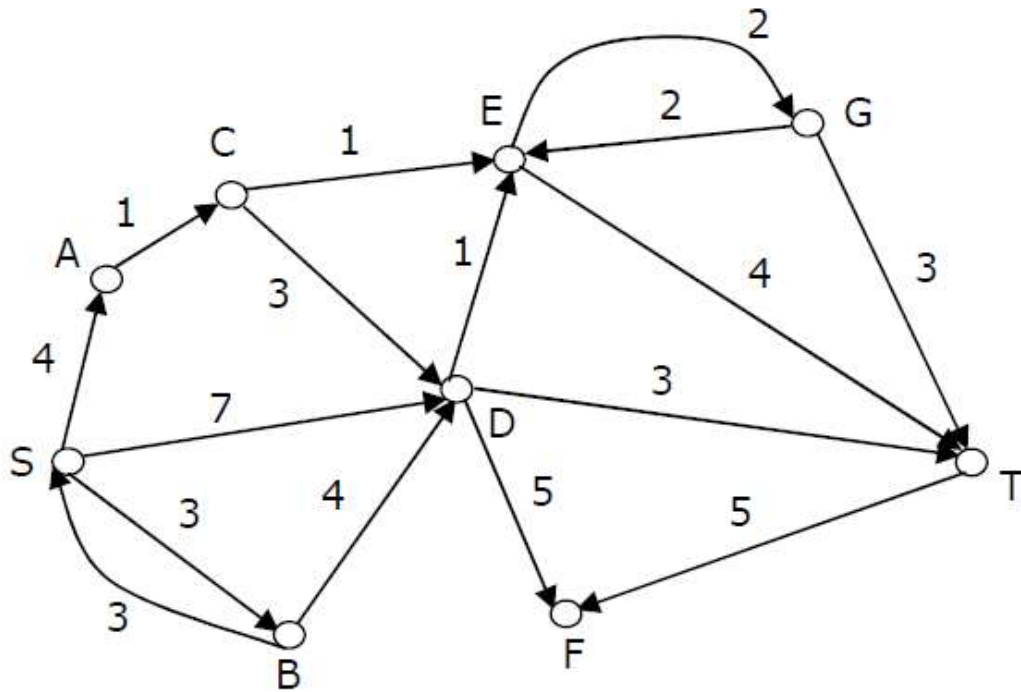
1/1

- ☐ $r=r+1$
- ☒ $r=(r+1)\% \text{QUEUE_SIZE}$
- ☐ $r=(r+1)\% \text{QUEUE_SIZE}-1$
- ☐ $r=(r-1)\% \text{QUEUE_SIZE}$



- ✗ 8. Consider the directed graph shown in the figure below. There are multiple shortest paths between vertices S and T. Which one will be reported by Dijkstra's shortest path algorithm? Assume that, in any iteration, the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered.

*0/1



- ☐ SDT
- ☐ SBDT
- ☒ SACDT
- ☐ SACET

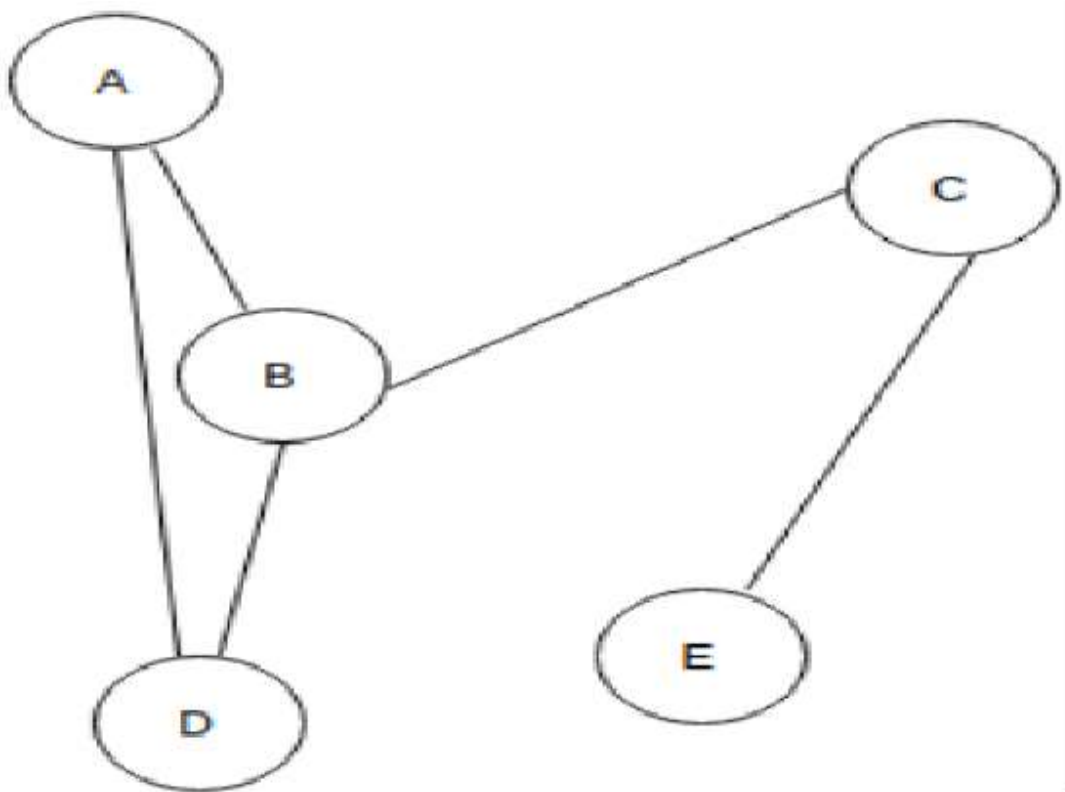
✗

Correct answer

- ☒ SACET

✗ 14. In the given graph identify the cut vertices *

0/1



- ☒ B and E
- ☐ C and D
- ☐ A and E
- ☐ C and B

Correct answer

- ☒ C and B

✗

✗ 15. Which one of the following is an application of Queue Data Structure? * 0/1

- ☐ When a resource is shared among multiple consumers
- ☐ When data is transferred asynchronously (data not necessarily received at same rate as sent) between two processes
- ☐ Load Balancing
- ☒ All of the above

✗

Correct answer

- ☒ When data is transferred asynchronously (data not necessarily received at same rate as sent) between two processes

✗ 34. choose the output
package demo;

*

0/1

```
public class DemoTranslation {  
    public static void main(String[] args) {  
        int i = lAbc(10);  
        System.out.println(--i);  
    }  
  
    public static int lAbc(int i) {  
        return i++;  
    }  
}
```

- ☒ 10
- ☐ 11
- ☐ 9
- ☐ 8

✗

Correct answer

- ☒ 9

- ✗ 36. Consider the following operation along with Enqueue and Dequeue operations on queues, where k is a global parameter. *0/1
- What is the worst case time complexity of a sequence of n MultiDequeue() operations on an initially empty queue?
- (A) $\Theta(n)$ (B) $\Theta(n + k)$ (C) $\Theta(nk)$ (D) $\Theta(n^2)$

```
MultiDequeue(Q){  
    m = k  
    while (Q is not empty and m > 0) {  
        Dequeue(Q)  
        m = m - 1  
    }  
}
```

- ☐ A
- ☐ B
- ☒ C
- ☐ D

✗

Correct answer

- ☒ A

✗ 19. What is the condition of the overflow of a linear queue implemented using an array? *0/1

- ☐ Rear = front + 1
- ☐ Rear = front
- ☒ Rear = MAX_SIZE
- ☐ Rear = MAX_SIZE - 1

✗

Correct answer

- ☒ Rear = MAX_SIZE - 1

✓ 37. Consider the following statements: *

1/1

- i. First-in-first out types of computations are efficiently supported by STACKS.
- ii. Implementing LISTS on linked lists is more efficient than implementing LISTS on an array for almost all the basic LIST operations.
- iii. Implementing QUEUES on a circular array is more efficient than implementing QUEUES on a linear array with two indices.
- iv. Last-in-first-out type of computations are efficiently supported by QUEUES.

- ☐ (ii) is true
- ☐ (i) and (ii) are true
- ☒ (iii) is true
- ☐ (ii) and (iv) are true

✓

✗ 26. The following formula is of *
 $\text{left_subtree}(\text{keys}) \leq \text{node}(\text{key}) \leq \text{right_subtree}(\text{keys})$

0/1

- ☐ Binary Tree
- ☒ Complete Binary Tree
- ☐ Binary Search tree
- ☐ All of the above

✗

Correct answer

- ☒ Binary Search tree

✗ 13. Find the Big O complexity of the following code: *

0/1

```
class NestedLoop {
    public static void main(String[] args) {
        int n = 10; // 1 step --> Note: n can be anything. This is just an example
        int sum = 0; // 1 step
        double pie = 3.14; // 1 step

        for (int var = 0; var < n; var = var + 3) { // n/3 steps
            System.out.println("Pie: " + pie); // n/3 steps
            for (int j = 0; j < n; j = j + 2) { // (n/3 * n/2) steps
                sum++; // (n/3 * n/2) steps
                System.out.println("Sum = " + sum); // (n/3 * n/2) steps
            }
        }
    }
}
```

- ☐ $O(n^2)$
- ☒ $O(n)$
- ☐ $n(0)$
- ☐ $O^2(n)$

✗

Correct answer

- ☒ $O(n^2)$

✓ 24. What will be the running-time of Dijkstra's single source shortest path algorithm, if the graph $G(V, E)$ is stored in form of adjacency list and binary heap is used

- ☐ $O(|V| \log |V|)$
- ☐ $O(|V|^2)$
- ☒ $O(|E| + |V| \log |V|)$
- ☐ None of these



✗ 9. A connected planar graph having 6 vertices, 7 edges contains _____ regions *0/1

- ☒ 15
- ☐ 7
- ☐ 3
- ☐ 1



Correct answer

- ☒ 3

✓ 4. Suppose the numbers 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are inserted in that order into an initially empty binary search tree. The binary search tree uses the usual ordering on natural numbers. What is the in-order traversal sequence of the resultant tree? 1/1

- ☐ 7510324689
- ☐ 0243165987
- ☒ 0123456789
- ☐ 9864230157



✗ 22. Which of the following is the time complexity if the graph is represented as an adjacency matrix in kruskal's algorithm?

*0/1

- ☐ $O(E \log V)$
- ☒ $O(V \log E)$
- ☐ $O(V^2)$
- ☐ $O(\log E)$

✗

Correct answer

- ☒ $O(E \log V)$

✓ 11 What does the following fragment of code do? *

1/1

```
Node currentNode = list.headNode;
while(currentNode != null){
    currentNode = currentNode.nextNode;
}
```

- ☒ Traverse
- ☐ sort
- ☐ Hash
- ☐ none of the above

✓

✓ 29. A graph is a tree if and only if graph is *

1/1

- ☐ Directed graph
- ☒ Contains no cycles
- ☐ Planar
- ☐ Completely connected



✓ form of access is used to add and remove nodes from a queue. *

1/1

- ☐ LIFO, Last In First Out
- ☒ FIFO, First In First Out
- ☐ Both a and b
- ☐ None of these



— — — .

